Supplementary Material: Theoretical Derivation and Validation on a Second Plant

Part A: From Euler–Bernoulli Beam Theory to the Relative Compliance Model

I. Overview and Goal

The main paper models tree-branch motion from video by asking: *which branches are more flexible than others?* To answer this, we need a mathematical relationship between (i) the bending we can measure from video and (ii) the stiffness and the wind load acting on each branch.

This annex derives that relationship starting from classical beam mechanics and ending at the simple product model

$$y_{e,t} \;\approx\; \kappa_e \, z_t, \tag{1}$$

where $y_{e,t}$ is the observed bending of branch segment (edge) e at video frame t, κ_e is the relative compliance of that segment, and z_t is a single number capturing how strong the wind is at frame t.

II. Step 1 : Modelling a Branch as a Beam

A. The standard beam equation

We treat each branch segment as a *slender beam*, a long, thin elastic rod that bends under a sideways (transverse) load. This is the Euler-Bernoulli beam model, the standard engineering approximation for slender structural members.

Let $s \in [0, L]$ be a coordinate measured along the branch from the joint at $s = 0$ to the free tip at $s = L$. Let $v(s, t)$ be how much the branch deflects sideways at position s and video frame t. Let $q(s, t)$ be the sideways wind force per unit length acting at position s and frame t.

The Euler–Bernoulli equation states

$$EI \, \frac{\partial^4 v}{\partial s^4} = q(s, t), \tag{2}$$

where EI is the *bending stiffness*: E is Young's modulus (how stiff the wood is) and I is the second moment of area (how the cross-section resists bending). In plain terms, Eq. (2) says: *the fourth derivative of the deflection shape equals the applied load divided by the stiffness*. A stiffer branch (large EI) deflects less under the same load.

B. Why the fourth-order form is inconvenient

Working with a fourth-order differential equation requires four boundary conditions and is awkward when we want to reason about forces and moments physically. We therefore convert it into an equivalent integral form involving the *bending moment*.

III. Step 2: Converting to the Moment Form

A. Shear force and bending moment

At any cross-section s along the branch:

- The *shear force* $V(s, t)$ is the net sideways force that everything to the right of s (i.e. the portion $[s, L]$) exerts on the left portion.
- The *bending moment* $M(s, t)$ is the net twisting action that the right portion exerts on the left portion.

Because the tip of the branch is free (nothing holds it), both the shear force and the bending moment are zero there:

$$V(L, t) = 0, \qquad M(L, t) = 0. \tag{3}$$

B. Deriving shear force by integration

The shear force at position s equals the total distributed load acting *beyond* s (between s and the tip). Integrating from s to L gives

$$V(s, t) \;=\; \int_s^L q(\xi,\, t)\, \mathrm{d}\xi. \tag{4}$$

Intuitively: if you cut the branch at s and hold the left piece, the right piece pulls on the cut with a total sideways force equal to all the wind load on that right piece.

C. Deriving bending moment: Cauchy's formula for repeated integration

Integrating the shear force (4) once more from s to L yields the bending moment as a *double* integral:

$$M(s, t) \;=\; \int_s^L V(\eta,\, t)\, \mathrm{d}\eta \;=\; \int_s^L \left(\int_\eta^L q(\xi,\, t)\, \mathrm{d}\xi \right) \mathrm{d}\eta. \tag{5}$$

Applying the *Cauchy formula for repeated integration* (the $n = 2$ case) collapses this into a single integral:

$$\boxed{M(s, t) \;=\; \int_s^L q(\xi,\, t)\, (\xi - s)\, \mathrm{d}\xi.} \tag{6}$$

The factor $(\xi - s)$ is the *lever arm*, the distance from the load location ξ to the observation point s. Equation (6) is therefore exactly *force times distance*, summed continuously along the branch.

IV. Step 3: Linking Moment to Curvature

Euler–Bernoulli theory also provides the *moment–curvature relation*:

$$EI \, \frac{\partial^2 v}{\partial s^2} = M(s, t). \tag{7}$$

The second derivative $\partial^2 v / \partial s^2$ is the curvature, how sharply the branch bends at s. Substituting Eq. (6) into Eq. (7) and dividing both sides by EI:

$$\frac{\partial^2 v}{\partial s^2} \;=\; \frac{1}{EI} \int_s^L q(\xi,\, t)\, (\xi - s)\, \mathrm{d}\xi. \tag{8}$$

Equation (8) is the central result of this section: *curvature is proportional to load and inversely proportional to stiffness*.

V. Step 4: Why Absolute Stiffness Cannot Be Recovered

From monocular video we can estimate branch *motion*, and from motion we can estimate the curvature on the left-hand side of Eq. (8). However, the wind load $q(s,t)$ is *not* measured.

This creates an irresolvable ambiguity. Replace EI by $c \cdot EI$ and q by $c \cdot q$ for any positive scalar c:

$$\frac{c \cdot q(\xi,t)}{c \cdot EI} = \frac{q(\xi,t)}{EI}. \tag{9}$$

The curvature is unchanged, so the video looks identical. In other words, a very stiff branch under a very strong wind looks exactly the same as a flexible branch under a gentle breeze. No amount of video analysis can distinguish these cases without an independent measurement of the load.

Conclusion: Absolute stiffness EI is not identifiable from monocular video alone.

VI. Step 5: What Is Identifiable: Relative Compliance

A. The shared-excitation assumption

Although absolute stiffness is unidentifiable, we can recover *relative* stiffness across branches if they are all blown by the same airflow. We model the load on branch segment e as a product of a *time-varying amplitude* $a_t \geq 0$ (how strong the wind is at frame t, the same for all branches) and a *fixed spatial profile* $\bar{q}_e(\xi)$ (how load is distributed along segment e):

$$q_e(\xi,t) \approx a_t \, \bar{q}_e(\xi). \tag{10}$$

This is called a *separable* load model. It is justified when all branches face the same upstream wind and each branch's aerodynamic shape stays roughly constant over time.

B. Substituting the separable load into the beam equation

Inserting Eq. (10) into Eq. (8) for edge e:

$$\frac{\partial^2 v_e}{\partial s^2} = a_t \cdot \frac{1}{EI_e} \underbrace{\int_s^{L_e} \bar{q}_e(\xi)\,(\xi - s)\,\mathrm{d}\xi}_{C_e(s)}. \tag{11}$$

The integral $C_e(s)$ depends only on the geometry and load shape of segment e, it does *not* depend on time. The time dependence is carried entirely by a_t.

C. From continuous curvature to a discrete bending signal

We cannot directly measure continuous curvature from video. Instead, for each segment e at each frame t, we compute a single bending angle $y_{e,t}$, for example, the change in the angle between two tracked skeleton points on the branch. This discrete measurement is a linear functional of the curvature field; applying it to Eq. (11) gives

$$y_{e,t} \approx a_t \cdot \frac{C_e}{EI_e}, \tag{12}$$

where C_e now absorbs the combined effect of the spatial load profile and the discrete angle measurement operation.

D. Defining relative compliance

We define the *relative compliance* of edge e as

$$\kappa_e \propto \frac{C_e}{EI_e}, \tag{13}$$

and we let z_t be proportional to the wind amplitude a_t. Equation (12) then becomes the *rank-1 observation model*:

$$\boxed{y_{e,t} \approx \kappa_e \, z_t.} \tag{14}$$

This is Eq. (1) from the overview. Its interpretation is simple:

- $y_{e,t}$: how much segment e bends in frame t (measured from video).
- z_t: how strong the wind is in frame t (shared by all segments, inferred jointly).
- κ_e: how compliant segment e is relative to the other segments (what we want to estimate).

A segment with $\kappa_e > 1$ bends more than average under the same wind and is therefore *more flexible* than average. A segment with $\kappa_e < 1$ bends less than average and is *stiffer* than average.

Part B: Validation on a Second Plant

VII. Motivation and Scope

The main paper demonstrates the relative compliance pipeline on a single specimen, hereafter *Plant 1*. As acknowledged in the limitations of the main paper, results from a single plant, with a single wind source, in a single session do not by themselves establish that the pipeline behaves consistently across specimens. To begin addressing this concern, we apply the same pipeline, with no algorithmic change and no per-plant tuning, to a second potted plant, hereafter *Plant 2*. This part reports the smoothing diagnostics, motion gradient, per-edge compliance map, projected overlays, uncertainty profile, shared excitation signal, and a zone-aggregated summary for Plant 2, and contrasts these against Plant 1 wherever a direct comparison is informative.

The goal of this validation is *not* to claim quantitative agreement between two plants; the two specimens differ in skeleton length, branching pattern, and leaf coverage, so the absolute κ_e values are not expected to match. The goal is to verify that the qualitative behaviours predicted by the rank-1 model are reproduced: monotonic root-to-tip motion amplification, bounded relative compliance values around the neutral level $\kappa = 1$, identifiable stiff and compliant zones, and a temporally structured shared excitation z_t that tracks the actual airflow.

VIII. Experimental Setup

Plant 2 is a different potted plant placed in front of an Intel RealSense D435i camera on a tripod and excited by the same constant-velocity airflow source used in the main paper. The camera is held in a fixed position so that no background motion contaminates the tracks, and only the monocular RGB

stream is used as input. The branch is annotated as a chain of 25 nodes with 24 edges in the first frame, and these anchors are tracked through 590 frames using TAPIR. As in the main paper, the tree is traversed from the root node to the most distal tip, with the root distance increasing monotonically from edge 1 to edge 24. Edges are assigned to three anatomical zones based on their distance from the root: *proximal* (edges 1–7, 7 edges), *mid* (edges 8–15, 8 edges), and *distal* (edges 16–24, 9 edges). Because each bending angle requires a grandparent–parent–child triplet, edge 1 has no measurable bending and the per-edge compliance is reported for the remaining 23 edges.

We use exactly the same hyperparameters as in the main paper: the raw TAPIR trajectories are first processed with a moving-average filter (window size 5) and then refined by the factor-graph smoother with $\lambda_m = 1.0$, $\lambda_v = 0.35$, $\lambda_a = 0.2$, $\lambda_e = 0.45$. Compliance is estimated by alternating least squares with shrinkage prior strength $\lambda = 12$, and bootstrap uncertainty is computed with block size $B = 25$ frames over 250 repetitions.

IX. Projected Compliance and Uncertainty Overlays

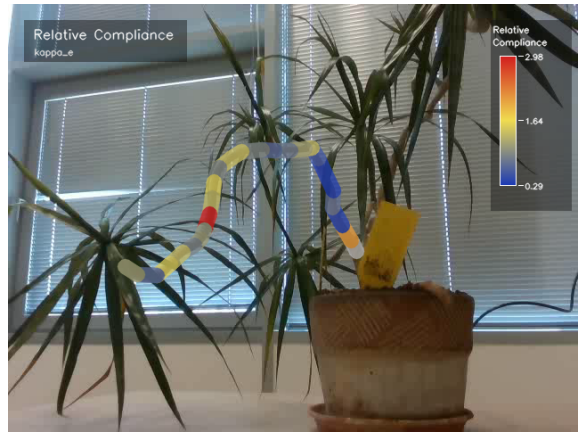
Before turning to per-edge plots, we present the same anatomical overlay used in the main paper, which projects the estimated κ_e and its bootstrap uncertainty U_e onto the first frame of the input video (Fig. 1). Plant 2's branch traces a long arc from the pot at the lower right, up and over to a small leafy plant at the upper left, with foliage occluding parts of the stem in the mid region.

In Fig. 1(a), three visual features are immediately apparent. First, a single bright red dot on the upper-left portion of the arc marks the most compliant edge in the entire skeleton, edge 17, with $\kappa_e = 2.98$. Second, a cluster of dark blue dots is concentrated near the pot at the lower right, identifying the proximal stiff region (edges 3, 5, and 6, all with $\kappa_e < 0.55$). Third, a small group of yellow/orange dots is scattered along the arc, marking edges of intermediate compliance. We discuss these features in detail in Sec. XII.

In Fig. 1(b), the uncertainty pattern is more diffuse. Most of the branch is rendered in yellow/orange (median $U_e \approx 0.43$), reflecting the moderate temporal evidence available in a 590-frame recording. A few green dots concentrated in the mid–distal arc indicate well-constrained estimates, while several reddish dots in the proximal cluster and around the occluded mid section flag edges where the bootstrap CIs are widest. The two overlays jointly summarise the same information that the per-edge plots in Fig. 4 present numerically.

X. Impact of Smoothing in Noise Reduction

The factor-graph smoother again removes a substantial portion of the geometric noise from the raw tracks while preserving the underlying bending signal (Fig. 2). Across the 23 non-root bends, the mean bending RMS decreases from $4.42°$ on the raw moving-averaged tracks to $2.11°$ on the smoothed tracks, a reduction of 52.2%. The reduction is larger than the 39.1% reported for Plant 1, which is expected: Plant 2's raw tracks contain spikes of up to $\sim 10°$ in the mid and distal zones,



(a)

(b)

Fig. 1: Projection of estimated κ_e and corresponding uncertainty as a heatmap on the first frame of the Plant 2 video. (a) Relative projected compliance: blue indicates stiff sections ($\kappa_e \to 0.29$) and red indicates the most compliant sections ($\kappa_e \to 2.98$). The single bright red dot on the upper-left arc corresponds to edge 17, the most compliant edge of the entire skeleton. (b) Relative uncertainty U_e of the same edges: green indicates well-constrained estimates ($U_e \to 0.22$), red indicates unreliable estimates ($U_e \to 0.62$). Most edges are in the moderate-uncertainty (yellow/orange) range expected for a 590-frame recording.

where the camera-plane motion is large and TAPIR's along-axis matches drift more aggressively. The smoother shrinks these spikes substantially without flattening the genuine bending signal in the proximal region, where motion is small and noise would otherwise dominate.

The edge-length coefficient of variation, $CV = \sigma/\mu$, exhibits a similar trend (Fig. 2b). Across all 24 edges, the mean CV decreases from 0.077 on the raw tracks to 0.044 on the smoothed tracks, a reduction of 42.8%. The reduction is most pronounced in the distal zone, where the raw CV reaches up to 0.20 and the smoothed CV stays in the range 0.04–0.11, indicating that the inextensibility regularizer λ_e stabilises edge length without overconstraining bending. The mean smoothed node motion (7.86 px) is essentially unchanged from the raw

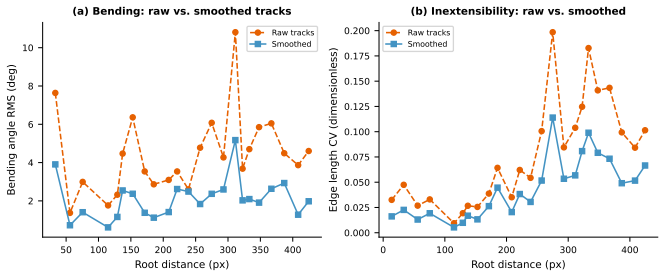


Fig. 2: Effect of factor-graph smoothing on Plant 2. Left: bending angle RMS across the branch as a function of root distance. Right: edge-length coefficient of variation. Smoothed tracks (blue) reduce the mean bending RMS by 52.2% and the mean edge-length CV by 42.8% compared to the raw moving-averaged tracks (orange), while leaving the overall motion signal essentially unchanged.

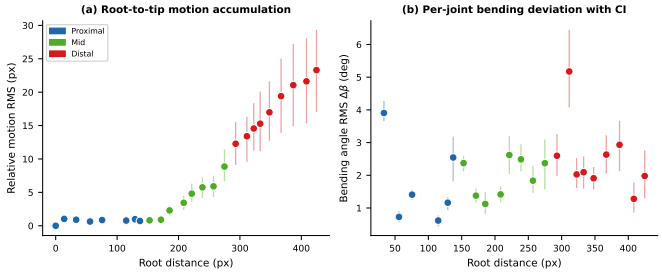


Fig. 3: Branch motion analysis for Plant 2. (a) Node relative motion RMS as a function of root distance, showing a near-monotonic root-to-tip amplification across all three anatomical zones. (b) Per-joint bending angle RMS $\Delta\beta$ with 95% bootstrap confidence intervals (vertical bars).

mean (8.02 px), confirming that the smoother acts on noise rather than on the underlying motion field. The smoother, in other words, pays a small budget in measurement fidelity to gain a much larger reduction in geometric noise, which is exactly what is needed before any compliance estimation can be reliable.

XI. Motion Amplification Along the Branch

Fig. 3(a) shows the relative motion RMS at each node as a function of root distance. The trend is markedly cleaner than in Plant 1: motion accumulates near-monotonically from the proximal zone (mean 0.74 px), through the mid zone (mean 4.10 px), to the distal zone (mean 17.55 px), with the most distal node reaching a relative motion RMS of 23.31 px. This is the textbook cantilever-beam response under distributed transverse load: each branch segment integrates the bending of all segments below it, so motion accumulates along arc length toward the tip. Plant 2 makes this lever-arm effect particularly clear because its skeleton is longer and the distal portion is less restrained by neighbouring foliage than in Plant 1.

The per-joint bending in Fig. 3(b), in contrast, is *not* a monotonic function of root distance. The largest bend occurs at root distance $\sim$311 px (edge 17, in the distal zone) with an RMS of $5.17°$, while several other bends, both proximal

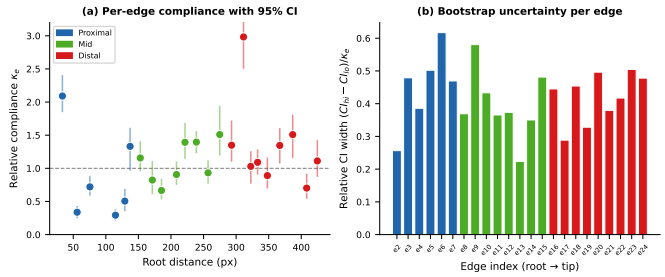


Fig. 4: Plant 2 relative compliance estimation. (a) κ_e per edge as a function of root distance; vertical bars are 95% moving-block bootstrap confidence intervals; the dashed line marks neutral compliance ($\kappa = 1$). Colours indicate anatomical zone. (b) Relative bootstrap CI width U_e per edge; edges with limited support or low signal-to-noise ratio show higher relative uncertainty.

and mid, sit between $0.6°$ and $1.5°$. This reproduces the same qualitative observation made in the main paper: branch deformation along the skeleton is heterogeneous, and which segments bend the most is determined by local material and geometric factors, not solely by the lever arm. This is precisely the structure the relative compliance proxy is designed to expose.

XII. Relative Projected Compliance Map

Fig. 4(a) shows the per-edge relative compliance κ_e for all 23 non-root edges with their 95% bootstrap confidence intervals. The most compliant edge is edge 17 in the distal zone, with $\kappa_e = 2.98$ and a bending RMS of $5.17°$, consistent with the per-joint bending peak in Fig. 3(b). This edge is well-supported (573 of 590 frames) and has a tight CI (relative width 0.29), so its high compliance is an authentic signal rather than a small-sample artifact. This observation anatomically corresponds to Fig. 1(a), where edge 17 appears as the single bright red dot on the upper-left segment of the arc. Other clearly compliant edges ($\kappa_e \gtrsim 1.5$) are edge 22 in the distal zone ($\kappa_e = 1.51$, bending RMS $2.93°$) and edge 15 in the mid zone ($\kappa_e = 1.51$, bending RMS $2.37°$); together with edge 17, these three edges define the locations along the branch where bending concentrates, and they correspond visually to the orange/red overlays distributed along the upper part of the arc in Fig. 1(a).

The stiffest section of the branch lies in the proximal zone: edges 5 ($\kappa_e = 0.29$), 3 ($\kappa_e = 0.34$), and 6 ($\kappa_e = 0.51$) all sit clearly below the neutral level. These edges form the dark-blue cluster visible near the pot in the lower-right portion of Fig. 1(a). They correspond anatomically to the part of the stem closest to the soil–pot interface, where the trunk is thickest and where the local geometry is closest to a clamped boundary. The recovered map therefore agrees with the prior expectation that the basal stem of a potted plant is the most rigid part of its skeleton. We note that, unlike Plant 1, where stiff sections were observed in the mid and distal regions, Plant 2 concentrates its stiff segments in the proximal zone; this difference reflects genuine anatomical variation between the two specimens rather than a methodological inconsistency.

One isolated proximal edge, edge 2 with $\kappa_e = 2.09$, deserves a separate note. It sits very close to the root (33 px) and yet shows above-average compliance with a relatively tight CI (relative width 0.26). It is visible as a yellow/orange dot near the pot in Fig. 1(a), distinct from the blue cluster around it. Two interpretations are plausible. First, the very basal triplet involves the fixed root anchor and the next two nodes, so any small rotation of the entire stem about the pot is registered as a bending of this single edge. Second, real potted plants are not perfectly clamped at the soil line; the stem–soil interface is itself slightly compliant. Both effects amplify the apparent compliance of the most basal joint, and both are properties of the experimental setup rather than of the stem material. We flag this as a known boundary effect rather than a failure of the estimator.

For robotic manipulation, the practical reading of Figs. 1(a) and 4(a) is the same as in the main paper. A gripping point that lies above (i.e., distal to) the most compliant edges places those edges between the contact point and the root, allowing them to absorb load like a spring rather than transmitting it rigidly. For Plant 2, this argues for grasping above edge 17, the bright red dot in Fig. 1(a), if the goal is to displace the distal portion of the branch with minimal stress on the proximal stem.

XIII. Uncertainty Analysis

The relative bootstrap CI width U_e across the 23 edges of Plant 2 has a median of 0.43 (43%), which is noticeably higher than the 29% reported for Plant 1. Two factors account for this. First, Plant 2 was recorded over 590 frames as opposed to the 1104 frames of Plant 1, so each edge has roughly half the temporal evidence available for bootstrap resampling. Second, several mid-zone edges of Plant 2 lose tracking for extended periods due to occlusion by leaves; for example, edges 9 and 10 carry only 428 inlier frames each. Both effects directly inflate bootstrap variability, which is the intended behaviour of the moving-block bootstrap as a stability diagnostic.

Of the 23 non-root edges, four have $U_e > 0.5$ and would be flagged as unreliable: edge 6 (proximal, $U_e = 0.62$), edge 9 (mid, $U_e = 0.58$), edge 23 (distal, $U_e = 0.51$), and edge 5 (proximal, $U_e = 0.50$). Two of these are proximal edges with intrinsically low bending RMS (1.16° and 0.62°, respectively), where the signal-to-noise ratio is poor regardless of frame count. The other two are edges with reduced track support: edge 9 has only $428/590$ inlier frames and edge 23 has $583/590$ but a low absolute bending RMS of 1.28°. These four edges are visible as the reddish dots in Fig. 1(b), and the per-edge bar plot in Fig. 4(b) makes the same ordering quantitatively explicit.

The log-space shrinkage prior pulls these weakly-supported edges toward $\kappa_e = 1$, and the bootstrap CIs widen accordingly. The combination of shrinkage and bootstrap therefore produces a self-consistent uncertainty story: where the data is informative, the estimate is sharp (e.g., edge 13 in the mid zone with $U_e = 0.22$, the lowest in the recording); where the data is poor, the estimate is conservatively pulled toward neutral and flagged as such. For a downstream robotic system, this is the desired behaviour: an edge with $U_e > 0.5$ is one that a manipulation planner should treat as uninformative rather than as a confident estimate of low compliance.

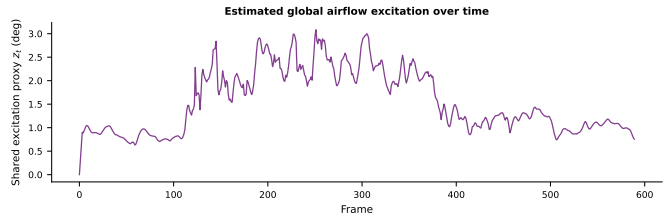


Fig. 5: Estimated shared excitation proxy z_t for Plant 2 over the 590-frame sequence, capturing the periodic airflow from the blower. The waveform is reconstructed jointly across all 23 edges by the alternating-least-squares update of the rank-1 model and has an RMS of 1.70°.

XIV. Shared Excitation Proxy

Fig. 5 shows the estimated shared excitation z_t for Plant 2. The waveform begins at zero in the first frame, rises to approximately 1° within the first ten frames, fluctuates around 0.8–1.0° through the first ∼100 frames, and then enters an extended sequence of broad gusts in which peaks reach up to 3.08°. The recorded RMS over the entire 590-frame sequence is 1.70°. As in the main paper, the consistency of the recovered z_t across edges is itself a non-trivial validation of the shared-excitation assumption: if different edges had been driven by uncorrelated stimuli, the alternating-least-squares update for z_t would not converge to a single coherent waveform. The fact that it does, on a second specimen with the same blower geometry, supports the model assumption articulated in Part A.

We note that the temporal structure of z_t for Plant 2 differs from Plant 1 in one respect: Plant 1's recording opened with a ∼8-second silent interval before the first gust (around frame 264), whereas Plant 2's recording captures airflow from frame 0 onwards. This is a property of the recording, not of the method.

XV. Zone-Aggregated Summary

The per-edge plots emphasise the heterogeneity of the branch, but a complementary view is obtained by aggregating each metric over its anatomical zone. Fig. 6 shows the proximal–mid–distal averages of three quantities: relative compliance κ_e, node relative motion RMS, and per-joint bending RMS $\Delta\beta$. All three quantities increase monotonically from proximal to distal, and the increases are coherent in direction even though the three quantities are constructed from different parts of the pipeline (geometry, dynamics, and the rank-1 model output, respectively).

The mean compliance crosses the neutral line between the proximal and mid zones: proximal segments are stiffer than the branch average ($\bar{\kappa} = 0.88$), the mid zone is approximately neutral ($\bar{\kappa} = 1.10$), and the distal zone is markedly more compliant ($\bar{\kappa} = 1.33$). This is exactly the qualitative ordering one would expect from a tapered woody stem under distributed wind load: the proximal stem is thicker and shorter-lever-armed, while the distal twigs are thinner and reach further from the

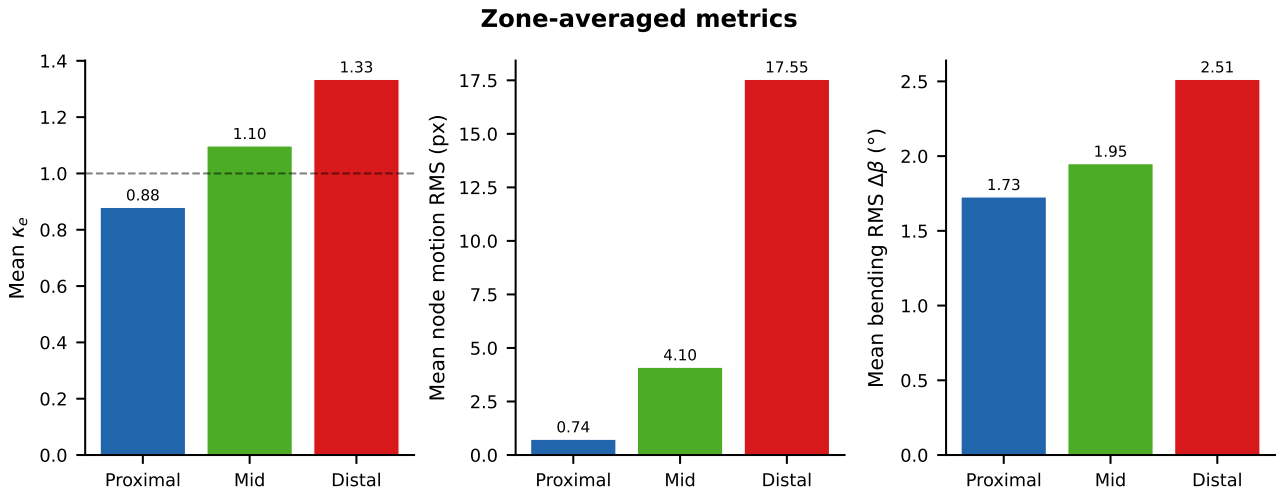


Fig. 6: Zone-aggregated metrics for Plant 2. Left: mean relative compliance κ_e per anatomical zone (dashed line marks neutral compliance $\kappa = 1$). Middle: mean node relative motion RMS in pixels. Right: mean per-joint bending RMS $\Delta\beta$ in degrees. All three quantities increase monotonically from proximal to distal, providing a coherent cross-check that the rank-1 compliance estimate, the geometric motion measurement, and the local angle measurement are mutually consistent.

TABLE I: Cross-plant comparison of headline pipeline outputs. Plant 1 numbers are reproduced from the main paper; Plant 2 numbers are computed in this supplement.

Quantity	Plant 1	Plant 2
Nodes / edges	20 / 19	25 / 24
Frames	1104	590
Bending RMS, raw	2.53°	4.42°
Bending RMS, smoothed	1.54°	2.11°
Smoothing reduction (RMS)	39.1%	52.2%
Edge-length CV (smoothed)	0.040	0.044
CV reduction	38.7%	42.8%
Median U_e	0.29	0.43
Most compliant edge κ_e	2.54 (e14)	2.98 (e17)
Min observed κ_e	–	0.29 (e5)
Excitation RMS	1.18°	1.70°

clamp. That this structural fact emerges automatically from a 25-node monocular video, with no force sensor and no per-plant calibration, is the core operational claim of the pipeline. Plant 2 reproduces it.

XVI. Cross-Plant Comparison and Discussion

Table I summarises the headline numbers from Plant 1 (main paper) and Plant 2 side by side. Three observations are worth highlighting.

The pipeline transfers without re-tuning

The same hyperparameter set used for Plant 1 is sufficient to produce a stable, interpretable compliance map for Plant 2, despite differences in skeleton size, branching pattern, frame count, and leaf occlusion. This is a useful sign that the pipeline is not over-fitted to a single specimen. It is, however, only one additional plant; multi-plant studies are still required for any quantitative cross-specimen claims.

Smoothing matters more when raw tracks are worse

Plant 2's raw tracks are noticeably noisier than Plant 1's, with raw bending RMS spikes reaching $\sim 10°$ in the distal zone. The factor-graph smoother absorbs more of this noise (a 52.2% reduction in mean bending RMS and a 42.8% reduction in mean edge-length CV, versus 39.1% and 38.7% for Plant 1) without distorting the underlying motion signal, as confirmed by the fact that the smoothed mean node motion (7.86 px) is essentially unchanged from the raw mean (8.02 px). In practice, the smoother is doing more useful work in the regimes where it is most needed.

Uncertainty is honest about reduced evidence

The median relative CI width U_e rises from 29% on Plant 1 to 43% on Plant 2. Rather than indicating that the method is less accurate on Plant 2, this rise is the moving-block bootstrap correctly reflecting that Plant 2 has roughly half the frames, suffers from more leaf-induced occlusion in the mid zone (edges 9 and 10 at $428/590$ frames), and contains lower-signal proximal edges. The combination of shrinkage and bootstrap reports these conditions transparently as widened CIs centred near $\kappa = 1$, rather than as overconfident point estimates. For a downstream robotic system, this is the desired behaviour: edges flagged with $U_e > 0.5$ are exactly the ones a manipulation planner should treat as uninformative.

XVII. Conclusion of the Validation

Across smoothing diagnostics, motion gradient, per-edge compliance, projected overlays, uncertainty profile, shared excitation, and zone-aggregated summaries, Plant 2 reproduces the qualitative behaviours predicted by the rank-1 observation model derived in Part A. Specifically, the pipeline recovers a near-monotonic root-to-tip motion gradient, identifies stiff and compliant segments along the branch (with edge 17 as

the dominant compliant outlier and the proximal cluster of edges 3, 5, and 6 as the stiffest section), and produces a temporally coherent shared excitation, all under the unchanged hyperparameters of the main paper. The widened bootstrap CIs for Plant 2 are consistent with the reduced frame count and the larger occluded fraction of this recording, and demonstrate that the uncertainty quantification adapts to the available evidence without manual intervention. Together with the main-paper experiment on Plant 1, this second specimen offers a first piece of evidence that the proposed pipeline behaves as expected on more than one specimen. A larger multi-plant study, with controlled variation in species, pot rigidity, and airflow geometry, is left for future work.